Example 3.3

The following table gives the random sample of marks scored by students in two schools, A and B.

School A	63	72	80	60	85	83	70	72	81
School B	86	93	64	82	81	75	86	63	63

Is the variance of the marks of students in school A is less than that of those in school B? Test at 5% level of significance.

Solution:

Let $X_1, X_2, \ldots, X_m$ represent sample values for school A and let $Y_1, Y_2, \ldots, Y_n$ represent sample values for school B.

Step 1 : **Null Hypothesis:** $H_1 : \sigma_X^2 = \sigma_Y^2$

That is, there is no significant difference between the two population variances.

Alternative Hypothesis: $H_1 : \sigma_X^2 < \sigma_Y^2$

That is, the variance of marks in school A is significantly less than that of school B.

Step 2 : **Data**

$X_1, X_2, \ldots, X_m$ are sample from school A

$Y_1, Y_2, \ldots, Y_n$ are sample from school B

Step 3 : **Test statistic**

$$F = \frac{s_X^2}{s_Y^2}$$

$$s_X^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{n-1}\sum_{j=1}^{n}(y_j - \bar{y})^2$$

Step 4 : **Calculations**

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
63	-11	121	86	9	81
72	-2	4	93	16	256
80	6	36	64	-13	169
60	-14	196	82	5	25
85	11	121	81	4	16
83	9	81	75	-2	4
70	-4	16	86	9	81
72	-2	4	63	-14	196
81	7	49	63	-14	196
666		628	693		1024

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